Explainable AI–Driven, Dynamic Climate-Aware Geospatial System for Wildfire Hazard and Community Risk Mapping
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Wildfire risk in California is intensifying. Prolonged drought, fuel buildup, and expanding development intersect across fire-prone regions. Existing hazard systems are often static, poorly validated, or evaluated under unrealistic conditions that inflate performance. Many omit social vulnerability, limiting their usefulness for equitable mitigation. I engineered a scalable, interpretable wildfire hazard and risk system designed to generalize to future fire seasons, handle extreme class imbalance, and connect environmental predictions to community vulnerability planning.

ClariFIRE predicts probabilistic wildfire hazard statewide by integrating terrain, vegetation indices, and seasonal climate and drought indicators. Through structured feature engineering including anomaly normalization, lagged fuel memory variables, and nonlinear interaction terms, the feature space expanded from 22 to 34 predictors. I evaluated each addition under strict validation to improve fire concentration in temporal and spatial testing without overfitting. Gradient boosting was selected based on superior ranking consistency and concentration performance.

The system was evaluated using unbalanced testing at real prevalence averaging 0.49%, temporal holdout with 2004–2020 training and 2022–2025 testing, and spatial cross-validation. Under realistic conditions, ROC-AUC reached 0.94. Under conservative temporal and spatial validation, ROC-AUC ranged from 0.77–0.81. Critically, in these validation regimes, the top 20% highest-risk areas captured 51–66% of wildfire events.

SHAP explainability was applied globally and at county levels. Hazard predictions were translated into community-level risk maps by integrating population exposure and CDC Social Vulnerability Index. ClariFIRE shows that wildfire risk, though rare, is spatially structured, predictable under realistic constraints, and actionable when integrated with social vulnerability. This enables equitable, data-driven mitigation planning.